

Supplementary information for: Heterogeneously Integrated Efficient and Widely Tunable Lasers at 795 nm for Rubidium-Based Quantum Technologies

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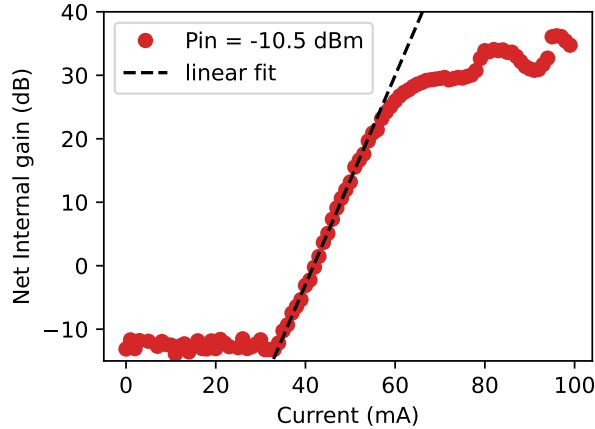
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SUPPLEMENTARY NOTE 1: REFLECTIVE AMPLIFIER NET GAIN

The net gain of the 1-mm-long Reflective Semiconductor Optical Amplifier (RSOA) coupon was measured in our previous work [1] and shown in Supplementary Fig. S1. More detail on the measurement setup can be found in the corresponding Supplementary Materials. The measured gain is net internal gain, meaning the double-pass gain without taking into account any loss from coupling into the RSOA waveguide.

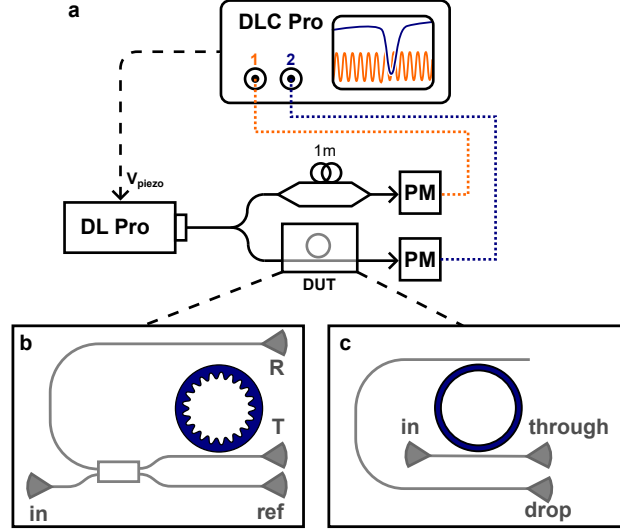


Supplementary Fig. S1: **Net internal gain of the RSOA.** Internal gain for different injection currents for a 1-mm-long RSOA measured at input power of -10.5 dBm in fiber (-16.8 dBm in RSOA), along with linear fit. The measurement floor of -12 dB is due to the ASE overpowering the low input signal power [1].

SUPPLEMENTARY NOTE 2: PASSIVE RESONATOR MEASUREMENT SETUP

The setup used to measure the microgear resonator and the MRRs is shown in Supplementary Fig. S2a. A Toptica DL Pro external-cavity diode laser (ECDL) operating around 780 nm is used. The Toptica DLC Pro laser controller is used to sweep the voltage of the internal piezo of the ECDL and to synchronously record the Power Meter (PM) output voltages. The diode laser output is split, with 90 % of the power going into the Device Under Test (DUT) and 10 % to a Mach-Zehnder interferometer where a 1 m extra fiber length is placed in one of the arms to act as a frequency reference. During analysis, this signal is used to convert the piezo voltage axis to frequencies. To measure the relevant

responses of the resonators used in this work, test structures were used with two distinct test benches on the photonic chip. The test bench for the microgear resonator is shown in Supplementary Fig. S2b and for the MRR in Supplementary Fig. S2c. To remove the multimode interferometer (MMI) loss from the microgear measurements, the cross coupling is measured from port ‘in’ to port ‘T’ and the bar coupling is measured from port ‘in’ to port ‘ref’. For the MRR measurement, the passband of the MRR through response is used as the amplitude reference. All responses are measured sequentially and the resulting spectra are overlaid by aligning the resonances.



Supplementary Fig. S2: **Schematic of the setup used to characterize the on-chip resonators.**

SUPPLEMENTARY NOTE 3: MICROGEAR REFLECTOR MODELING

To extract the physical parameters of the microgear reflectors from the experimental transmission and reflection spectra, an analytical model based on temporal coupled-mode theory (t-CMT) was employed [2]. This model is used (instead of a transfer-matrix model) as it can more accurately extract the quality factors of the underlying resonances and more directly extracts the coupling and loss rates in the system. Unfortunately, this model can only simulate one FSR of the resonator simultaneously. However, this fits well with the limited frequency range of the measurement.

In an ideal all-pass microring resonator, light circulates in a single direction. However, surface roughness and localized defects introduce backscattering, lifting the degeneracy of the clockwise (CW) and counter-clockwise (CCW) modes and coupling them together.

Following the theoretical framework for microring backscattering, this mutual coupling is represented by a lumped backscattering rate μ . The time-evolution of the clockwise (a_{cw}) and counter-clockwise (a_{ccw}) mode amplitudes within the microgear, driven by an input field s_{in} from the single bus waveguide, is described by the following coupled differential equations:

$$\frac{da_{cw}}{dt} = (i\omega_0 - \gamma_t)a_{cw} + i\mu a_{ccw} + i\sqrt{2\gamma_e}s_{in} \quad (1)$$

$$\frac{da_{ccw}}{dt} = (i\omega_0 - \gamma_t)a_{ccw} + i\mu a_{cw} \quad (2)$$

where ω_0 is the unperturbed angular resonance frequency, γ_e is the extrinsic coupling rate to the bus waveguide, γ_{in} is the intrinsic loss rate of the cavity, and $\gamma_t = \gamma_e + \gamma_{in}$ is the total decay rate. The parameter μ quantifies the strength of the internal backscattering that leads to resonance splitting. Throughout this note, and in all fitted values quoted in the main text, the rates γ_e , γ_{in} , γ_t and μ are expressed as ordinary frequencies (in Hz) rather than angular frequencies, so that the loaded quality factor follows directly as $Q_L = f_0/(2\gamma_t)$.

By assuming a harmonic time dependence $e^{i\omega t}$ and solving for the steady-state fields, the transmission (T) and reflection (R) transfer functions can be expressed as a function of the frequency detuning $\Delta f = f - f_0$:

$$T(\Delta f) = G_t \left| 1 - \frac{2\gamma_e(i\Delta f + \gamma_t)}{(i\Delta f + \gamma_t)^2 + \mu^2} \right|^2 \quad (3)$$

$$R(\Delta f) = G_r \left| \frac{i2\gamma_e\mu}{(i\Delta f + \gamma_t)^2 + \mu^2} \right|^2 \quad (4)$$

where G_t and G_r are experimental scaling factors applied to the transmission and reflection measurements, respectively.

This formulation assumes that the spatial phase shift between the CW and CCW modes is negligible ($\phi_\mu \approx 0$), resulting in symmetrically split doublets. This accurately captures the steady-state narrowband behavior of the measured high- Q resonances, allowing for the direct extraction of the loaded, intrinsic, and coupling quality factors.

SUPPLEMENTARY NOTE 4: ADD-DROP RESONATOR TRANSFER-MATRIX MODEL

For characterization of microring resonators in an add-drop configuration, the spectral transmission was modeled by assuming continuous-wave operation and matching the optical fields at the coupling junctions [3]. This framework yields an exact analytical description of the transmission to both the pass (through) and drop ports.

Assuming that counter-directional reflections back into the bus waveguide are negligible, the intensity transmission to the through/pass port (T_p) and the drop port (T_d) relative to the input intensity (I_{input}) are given by:

$$T_p = \frac{I_{pass}}{I_{input}} = \frac{r_2^2 a^2 - 2r_1 r_2 a \cos \phi + r_1^2}{1 - 2r_1 r_2 a \cos \phi + (r_1 r_2 a)^2} \quad (5)$$

$$T_d = \frac{I_{drop}}{I_{input}} = \frac{(1 - r_1^2)(1 - r_2^2)a}{1 - 2r_1 r_2 a \cos \phi + (r_1 r_2 a)^2} \quad (6)$$

where r_1 and r_2 represent the self-coupling coefficients of the input and output directional couplers, respectively. For the ring resonators in this work, $r_1 = r_2$ was assumed. Following the convention of [3], the parameter a defines the round-trip amplitude transmission of the cavity, which aggregates both the propagation losses within the ring and the insertion losses of the coupling sections. The accumulated round-trip phase shift is correspondingly denoted by $\phi = \beta L$, where β is the propagation constant and L is the round-trip length of the resonator.

When the attenuation is negligible ($a \approx 1$), critical coupling is achieved through a symmetric coupler design where the cross-coupling coefficients are equal ($k_1 = k_2$). For a lossy resonator, the critical coupling condition shifts such that the losses must match the coupling parameters, defined as $r_2 a = r_1$. However, this equation implies that critical coupling in both directions is impossible in presence of loss. To maximize coupling instead, the coupling must dominate the intrinsic loss ($1 - r^2 \gg 1 - a^2$), which deteriorates the loaded Q-factor of the ring. This relationship uncovers a trade-off in rings where losses are significant.

- [1] Kiewiet, M. et al. Micro-transfer printed continuous-wave and mode-locked laser integration at 800 nm on a silicon nitride platform. *Laser & Photonics Reviews* 20, e00956 (2026).
- [2] Li, A., Van Vaerenbergh, T., De Heyn, P., Bienstman, P. & Bogaerts, W. Backscattering in silicon microring resonators: a quantitative analysis. *Laser & Photonics Reviews* 10, 420–431 (2016).
- [3] Bogaerts, W. et al. Silicon microring resonators. *Laser & Photonics Reviews* 6, 47–73 (2012).